

interface for ease of use. It is also designed with a very nice plug in setup for new port-scanning tests. For more information, take a look at: <http://www.nessus.org>

Denial of Service Attacks

Denial of service attacks have increased greatly in recent years. Some of the more popular and recent ones are listed below. Note that new ones show up all the time, so this is just a few examples. Read the Linux security lists and the bugtraq list and archives for more current information.

- SYN Flooding - SYN flooding is a network denial of service attack. It takes advantage of a "loophole" in the way TCP connections are created. The newer Linux kernels (2.0.30 and up) have several configurable options to prevent SYN flood attacks from denying people access to your machine or services. See Section 7 for proper kernel protection options.
- Pentium "F00F" Bug - It was recently discovered that a series of assembly codes sent to a genuine Intel Pentium processor would reboot the machine. This affects every machine with a Pentium processor (not clones, not Pentium Pro or PII), no matter what operating system it's running. Linux kernels 2.0.32 and up contain a work around for this bug, preventing it from locking your machine. Kernel 2.0.33 has an improved version of the kernel fix, and is suggested over 2.0.32. If you are running on a Pentium, you should upgrade now!
- Ping Flooding - Ping flooding is a simple brute-force denial of service attack. The attacker sends a "flood" of ICMP packets to your machine. If they are doing this from a host with better bandwidth than yours, your machine will be unable to send anything on the network. A variation on this attack, called "smurfing", sends ICMP packets to a host with your machine's return IP, allowing them to flood you less detectably. You can find more information about the "smurf" attack at <http://www.quadrunner.com/~chuegen/smurf.txt>